

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 6

Time & Work

- Q1** 15 workers can complete $\frac{2}{3}$ of a project in 8 hours. How many hours will it take 6 workers to finish $\frac{5}{6}$ of the same project?
(A) 20 (B) 24
(C) 25 (D) 30
- Q2** A pipe can fill $\frac{3}{7}$ of a tank in 9 minutes. Another pipe can empty $\frac{2}{7}$ of the same tank in 8 minutes. If both pipes are opened together, how much time will it take to fill $\frac{5}{7}$ of the tank?
(A) 45 minutes
(B) 54 minutes
(C) 60 minutes
(D) None of the above
- Q3** A water tank, which normally gets filled in 12 minutes by an inlet pipe, took 15 minutes today due to a small leak at the bottom. If the inlet pipe is turned off once the tank is full, how much time will the leak take to completely empty the filled tank?
(A) 40 minutes (B) 45 minutes
(C) 54 minutes (D) 60 minutes
- Q4** Ravi needs 6 days more than Suraj to finish a task when working alone. If they work together, the task is completed in 4 days. Determine how many days Ravi would take to complete the task independently.
(A) 2 days (B) 6 days
(C) 10 days (D) 12 days
- Q5** Pump X can fill an entire tank in 6 hours, while pump Y can do the same job in 8 hours. Both pumps are turned on simultaneously, but pump X is stopped after a certain duration. If the tank gets completely filled in 4 hours, after how many hours was pump X turned off?
(A) 1 (B) 2
(C) 2.5 (D) 3
- Q6** While working alone, M can complete a piece of work in 12 days, N in 15 days and O in 20 days. Initially, all the three started working together but O left midway. The remaining work was completed by M and N together. If the whole work was completed in 6 days, then after how many days did O leave?
(A) 1 (B) 2
(C) 3 (D) 4
- Q7** Aisha can do a work in 30 days, while Piranha can do the same work in 45 days. Aisha started the work and was joined by Piranha after 5 days. At the end of 10th day from the beginning, they had an argument and Aisha quit. How many more days does Piranha require to complete the remaining work?
(A) 15 (B) 20
(C) 25 (D) 35
- Q8** A piece of work can be done by 3 men A, B and C in 10, 15 and 20 hours respectively. If A starts the work, then after 'x' hours, B joins, and once again after 'x' more hours, C joins, the work will be done in 5 hours from the beginning. Find the time (in minutes) after which B joined the work.
(A) 10 (B) 20
(C) 30 (D) 60
- Q9** Roger completes $\frac{3}{5}$ of a project on quantum mechanics in 3 days. Steve can complete $\frac{1}{5}$ of



the same project in 6 days and Barton can complete $\frac{5}{6}$ of the project in 15 days. All of them worked together for 3 days and then Roger and Barton left. How long will it take for Steve to complete the remaining project alone?

- (A) 3 days (B) 4 days
(C) 5 days (D) 6 days

Q10 A can do a work in 25 days and B in 50 days. If they work on it together for 6 days, then the fraction of the work that would still be left is

- (A) $\frac{9}{25}$ (B) $\frac{1}{10}$
(C) $\frac{7}{15}$ (D) $\frac{16}{25}$

Q11 Vijay can do a certain work in 12 days and Kailash can do the same job in 18 days. With the help of Chaman, all of them together completed the job in 6 days. Then, Chaman alone can do the job in

- (A) 24 days (B) 40 days
(C) 36 days (D) 10 days

Q12 Jyotsna can do a piece of work in 12 hours; Vishakha and Muskan together can do it in 2 hours, while Jyotsna and Muskan together can do it in 3 hours. How long will Vishakha alone take to do it?

- (A) 3 hours (B) 10 hours
(C) 4 hours (D) 12 hours

Q13 Barban is four times as good a workman as Alice and can build a wooden house in 45 days less than the number of days required by Alice. How many days will they take, working together, to build three such wooden houses?

- (A) 24 (B) 36
(C) 12 (D) 18

Q14 Rita, Gita, and Sita individually take different amounts of time to complete a task, with their times being in the ratio 4:6:9. Together, they can finish the task in 5 days, working 6 hours per day. However, Rita and Gita decide to

collaborate for the first 4 days, working 8:30 hours each day. How many hours will Sita require to complete the remaining task working alone?

Q15 A can finish 75% of a job in 12 days. B's efficiency is 20% greater than that of A. Both A and B worked together on the task for 5 days before stopping. C then completed the rest of the task in 20 days. So what percent more/less than that of A is C's efficiency?

Q16 If 3 men take the same time to complete a task as 4 women, and 2 women take the same time to finish the task as 6 children, a team of 3 men, 5 women, and 7 children can complete the task in 24 days. In how many days can a team of 3 men, 2 women, and 6 children complete the same task?

- (A) 24 (B) 30
(C) 32 (D) 34

Q17 X, Y, and Z can individually complete a certain task in 50, 70, and 90 days, respectively. They all start working on the task simultaneously. However, after a specific number of days, X takes a 3-day break. At the start of the 15th day, Y stops working entirely and does not return to the task. Furthermore, Z does not contribute on the final day when the task is completed. Considering these interruptions, determine the total number of days in which the task got completed.

- (A) 28 (B) 30
(C) 31 (D) 32

Q18 12 men started a job and worked on it for 12 days. At the start of the 13th day and onwards, a man drops out of the job everyday until it is finished by the last man working alone on the 23rd day. The same job is instead done by employing one man on the 1st day, and a new man joining each day from the 2nd day onwards. How many days will it take to finish the job in such a manner?



- (A) 22 (B) 25
(C) 15 (D) 20

- Q19** Babli can finish a job in 75 days, if she works alone. Abli is twice as fast as Babli but only one-third as fast as Chabli in the same job. Suppose Abli and Babli work together on the first day, Babli and Chabli work together on the second day, Chabli and Abli work together on the third day, and then, they continue the work by repeating this three-day roaster, with Abli and Babli working together on the fourth day, and so on. Then, the total number of days Babli would have worked when the job gets finished is
(A) 7 (B) 8
(C) 9 (D) 13
- Q20** Vikram and Dhanush can make a furniture in 9 days and 16 days respectively more than the time taken if both of them worked together. Find the time (in days) in which Vikram can make the furniture alone.
- Q21** The time taken by 5 men and 8 machines to finish a job together is 60% less than the time required by 4 men and 3 machines to complete the same task. If 8 machines can complete the job in 3 days, how many days will it take for 4 men to complete the job alone?
(A) 12 days (B) 24 days
(C) 48 days (D) 60 days
- Q22** Neha and Nisha were given a task that could only be finished on time if both Neha and Nisha worked together every day. However, the task was completed 7 days late due to their absences. Neha missed 6 days of work during the project, and Nisha missed 9 days. Determine the ratio of Neha's efficiency to that of Nisha's efficiency.
(A) 1:2 (B) 2:1
(C) 2:3 (D) 3:2

- Q23** Construction of a house can be completed in 16 days by 20 workers. If 25 workers started working from 1st of July, at the end of which day should 15 workers stop working such that the work gets completed on 20th of July?
(A) 6 July (B) 8 July
(C) 9 July (D) 10 July
- Q24** A and B digging together can dig a well in 15 days, B and C digging together will take 24 days for the same whereas A and C digging together will require 30 days to finish the work. If all three of them together were hired to finish the job and were paid Rs 680 at the end of the work, what will be B's share?
(A) Rs 360 (B) Rs 320
(C) Rs 280 (D) Rs 240
- Q25** P, Q, R, and S are two pipes and two leaks, respectively. Pipes P and Q can fill an empty tank in 4 hours and 6 hours, respectively, when working alone. Leaks R and S can empty a full tank in 8 hours and 12 hours, respectively. Starting with an empty tank, P is opened for the first 15 minutes and then closed. Next, R is opened for the next 15 minutes and then closed. Then, Q is opened for the following 15 minutes before being shut, and finally, S is opened for the last 15 minutes of the hour and then closed. This cycle repeats every hour in the same sequence until the tank is completely filled. How many minutes will it take in total to fill the tank?
- Q26** Three painters, David, John and Rudra, are tasked with painting a house. John requires 9 more hours than David to finish the painting job. Rudra alone can destroy all the work in 6 hours. When all three work together, they finish the job 8 hours faster than John would have, had he worked alone. How many hours will David, Rudra and John need, working together, to paint a house twice as large as the first one?



Q27 Ravi, Sanjay, and Priya collaborate on a construction project. On the first day, Ravi and Sanjay work together; on the second day, Ravi and Priya work; and on the third day, Sanjay and Priya work. They repeat this cycle and finish the project in exactly 72 days. Ravi alone could have completed the project in 96 days, while Sanjay's efficiency is 50% greater than Priya's. How many days would it take Sanjay to complete half the project on his own?

Q28 Two types of ants: Formica and Dorylus work on dragging a dead fly from point A to point B. 6 Formica and 10 Dorylus ants together can complete the dragging work in 6 minutes. If one Formica ant can complete the dragging work alone in 1 hour, then 1 Dorylus can complete the work alone in how many minutes?

Q29 Three inlet pipes A, B and C are connected to a tank. The time Pipes A and B together take to

fill the tank is the same as pipe C takes alone. Pipe B fills the tank 9 hours faster than pipe A and 3 hours slower than pipe C. If all the pipes are simultaneously opened and a leak L is created at the bottom of the tank, it takes 24 hours to fill the tank. Then, how much time (in the nearest integer hours) will it take to drain the tank by the leak alone?

- (A) 3 (B) 4
(C) 5 (D) 6

Q30 Ravi committed to finishing a project in 75 days. Initially, he used a certain number of workers but after 30 days, only 30% of the work was done. He then increased the workforce by 10 workers, and in the following 15 days, only an additional 20% of the work was completed. For the final 30 days, Ravi added 'n' more workers to complete the project on schedule. What will be the value of n given that the efficiency of all the workers was the same?



Answer Key

Q1 C
Q2 C
Q3 D
Q4 D
Q5 D
Q6 B
Q7 C
Q8 C
Q9 B
Q10 D
Q11 C
Q12 C
Q13 B
Q14 15
Q15 75

Q16 D
Q17 A
Q18 D
Q19 C
Q20 21
Q21 D
Q22 B
Q23 B
Q24 A
Q25 1095
Q26 8
Q27 80
Q28 150
Q29 A
Q30 10



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$$8 \times 15 = \frac{2}{3} \text{ of work}$$

$$1 \text{ worker in 1 hour} = \frac{2}{3 \times 8 \times 15} = \frac{1}{180}^{\text{th}}$$

of the work

$$\text{So worker hours required to complete the work} = 180$$

$$\text{So hours required for 6 workers to finish } \frac{5}{6}^{\text{th}} \text{ of the project} = \left(\frac{5}{6} \times 180 \right)$$

$$\div 6 = 25$$

Video Solution:



Q2 Text Solution:

$$\text{Tank filled in a minute by the pipe filling the tank} = \frac{3}{7 \times 9} = \frac{1}{21}$$

$$\text{Tank emptied in a minute by the pipe emptying the tank} = \frac{2}{7 \times 8} = \frac{1}{28}$$

$$\text{Tank filled in a minute when both the pipes are opened together} = \frac{1}{21}$$

$$- \frac{1}{28} = \frac{1}{84}$$

$$\text{Time required to fill } \frac{5}{7} \text{ of the tank when both the pipes are open} = \frac{5}{7} \times 84 = 60 \text{ minutes}$$

Video Solution:



Q3 Text Solution:

The tank is filled normally in 12 minutes but due to leak it took 15 minutes to fill. Hence 3 minutes of inlet = 15 minutes of leak
Hence to completely empty the tank the leak should remove 12 minutes of inlet pipe's work, so

$$\text{Time required for leak to empty the filled tank completely} = \frac{12}{3} \times 15 = 60 \text{ minutes}$$

Video Solution:



Q4 Text Solution:

Let the number of days required by Ravi to finish the work be x days.
So time required by Suraj to finish the task = $(x - 6)$ days

So

$$\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4}$$

$$\Rightarrow \frac{2x-6}{x^2-6x} = \frac{1}{4}$$

$$\Rightarrow x^2 - 6x - 8x + 24 = 0$$

$$\Rightarrow x^2 - 14x + 24 = 0$$

$$\Rightarrow (x - 12)(x - 2) = 0$$

$$\Rightarrow x = 12 \text{ or } x = 2$$

Since number of days cannot be negative, so $x = 12$ (or else $x - 6 = -4$)

So time required by Ravi to finish the task = 12 days

Video Solution:



Android App

iOS App

PW Website

**Q5 Text Solution:**

Taking volume of tank to be =

$$\text{LCM}(6, 8) = 24 \text{ l}$$

Filling rate of X = 4 l/h

Filling rate of Y = 3 l/h

Lets the number of hours be h for which both the pumps are working together, so

$$(4 + 3)h + 3(4 - h) = 24$$

$$\Rightarrow 7h + 12 - 3h = 24$$

$$\Rightarrow 4h = 12$$

$$\Rightarrow h = 3$$

So pump X is turned off after 3 hours.

Video Solution:**Q6 Text Solution:**

In one day, work done by M, N and O is $\frac{1}{12}$, $\frac{1}{15}$

and $\frac{1}{20}$ respectively.

Let O worked for X days.

$$\Rightarrow \frac{6}{12} + \frac{6}{15} + \frac{X}{20} = 1$$

$$\Rightarrow \frac{X}{20} = 1 - \frac{6}{12} - \frac{6}{15}$$

$$\Rightarrow \frac{X}{20} = \frac{1}{10}$$

$$\Rightarrow X = 2$$

After 2 days O left.

Video Solution:**Q7 Text Solution:**

Let the total work be 90 units

$$\text{Aisha} = \frac{90}{30} = 3 \text{ units per day}$$

$$\text{Piranha} = \frac{90}{45} = 2 \text{ units per day}$$

Work completed by Aisha in the first 10 days = $3 \times 10 = 30$ units

Work completed by Piranha in 5 days (She did not work for the first 5 days) = $2 \times 5 = 10$ units

Total work done = $30 + 10 = 40$ units

Remaining work = $90 - 40 = 50$ units

Time required by Piranha to complete the remaining work

$$= \frac{50}{2}$$

$$= 25 \text{ days.}$$

Video Solution:**Q8 Text Solution:**

Let us say the gap between the joining time of any two workers be x hours

Therefore, the first man worked for whole 5 hours,

2nd man worked for $(5 - x)$ hours, and

3rd men worked for $(5 - 2x)$ hours

By using the above conclusions, we can form this equation:

$$\frac{5}{10} + \frac{5-x}{15} + \frac{5-2x}{20} = 1$$

$$\Rightarrow \frac{30+20-4x+15-6x}{60} = 1$$

$$\Rightarrow \frac{65-10x}{60} = 1$$

$$\Rightarrow x = \frac{1}{2}$$



By solving this we get $x = \frac{1}{2}$ hours, i.e., 30 mins.

Video Solution:



Q9 Text Solution:

Roger does $\frac{3}{5}$ of the work in 3 days.

Roger takes 5 days to complete the whole work.

Part of work finished by Roger in one day = $\frac{1}{5}$

Steve does $\frac{1}{5}$ of the work in 6 days.

Steve takes 30 days to complete the work

Therefore, part of work finished by Steve in

one day = $\frac{1}{30}$

Barton does $\frac{5}{6}$ of the work in 15 days.

Barton takes 18 days to complete the work

Therefore, part of work finished by Barton in

one day = $\frac{1}{18}$

In 3 days, work done by all three would be:

$$= 3 \times \left(\frac{1}{5} + \frac{1}{30} + \frac{1}{18} \right)$$

$$= 3 \times \frac{13}{45}$$

$$= \frac{13}{15}$$

Therefore, the remaining work = $1 - \frac{13}{15} = \frac{2}{15}$

Work left for Steve would be $\frac{2}{15}$ of the work.

$$\text{Steve would require} = \frac{\frac{2}{15}}{\frac{1}{30}} = \frac{2}{15} \times 30 = 4$$

days.

Video Solution:



Q10 Text Solution:

Let the total work be the LCM of time taken which is 100 units.

Therefore in 1 day, A will complete $100/25 = 4$ units and B will complete $100/50 = 2$ units.

Therefore, both of them together will complete $4+2 = 6$ units in one day.

Thus, in 6 days, 36 units of work will be completed.

Therefore remaining work will be $100 - 36 = 64$ units.

Thus, fraction of work left is $64/100 = 16/25$.

Video Solution:



Q11 Text Solution:

Let total work be the LCM of 12 and 18 which is equal to 36 units.

Thus, in 1 day Vijay can complete $36/12 = 3$ units and Kailash can complete $36/18 = 2$ units.

Thus, 1 day's work of Vijay and Kailash will be $3+2 = 5$ units.

But with the help of Chaman, they could do the same in 6 days.

Thus, work completed per day would have been $36/6 = 6$ units.

Thus, in 1 day Chaman can complete $6 - 5 = 1$ units.

Thus Chaman will require a total of $36/1 = 36$ days to finish the work.

Video Solution:



Q12 Text Solution:



Let the total work be taken as LCM of 3, 4 and 12 i.e. 12 units.

Thus, Jyotsna will complete $12/12 = 1$ unit in an hour.

Vishakha and Muskan will complete $12/2 = 6$ units in an hour.

Further Jyotsna and Muskan will complete $12/3 = 4$ units in an hour.

This means that Muskan will complete 3 units in an hour.

Thus, Vishakha can do 3 units in an hour.

This means that Vishakha alone can do it in $12/3 = 4$ hours

Video Solution:



Q13 Text Solution:

As the ratio of efficiencies of Barban and Alice is 4:1, therefore the time taken by them will be in the ratio 1:4.

Let Barban take x days, then Alice will take $4x$ days.

$$\text{Thus, } 4x - x = 45$$

$$3x = 45$$

$$x = 15$$

Thus, Alice takes 60 days and Barban takes 15 days.

Therefore in one day they will complete $1/60 + 1/15$ part of work which is equal to $5/60$ or $1/12$

Therefore, a wooden house can be built by them in 12 days, thus they will require a total of 36 days to build three wooden houses.

Video Solution:



Q14 Text Solution:

	Rita	Gita	Sita
Time required	4	6	9
Ratio			
Work Rate Ratio	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{9}$
Integral work rate Ratio	9	6	4

Let the work rate of the workers be

$$\text{Rita} = 9k \text{ units/h}$$

$$\text{Gita} = 6k \text{ units/h}$$

$$\text{Sita} = 4k \text{ units/h}$$

where k is a constant.

So work done in one hour by all three = $19k$ units

So total work required to finish the task =

$$5 \times 6 \times 19k = 570k \text{ units}$$

So

work done by Gita and Rita initially =

$$15k \times 8.5 \times 4 = 510k \text{ units}$$

So remaining work to be done by Sita = $60k$ units

Time required by Sita to complete the work =

$$\frac{60k}{4k} = 15 \text{ hours}$$

Video Solution:



Q15 Text Solution:

A can finish the complete job in =

$$\frac{100}{75} \times 12 = 16 \text{ days}$$

B's efficiency as compared to A = 1.2A



So, number of days required by B to finish the

$$\text{job} = \frac{16}{1.2} = \frac{40}{3} \text{ days}$$

Let the total work be 80 units.

Efficiency of A = 5 units/day

Efficiency of B = 6 units/day

Amount of work done by A and B together in 5

days = 55 units

So work remaining = 25 units

$$\text{So C's efficiency} = \frac{25}{20} = 1.25 \text{ units/day}$$

So, C's efficiency is less than that of A's by =

$$\frac{5-1.25}{5} \times 100 = 75\%$$

Video Solution:



Q16 Text Solution:

$$3M = 4W$$

$$\Rightarrow M : W = 4 : 3$$

$$2W = 6C$$

$$\Rightarrow W = 3C$$

So

$$M : C = 4 : 1$$

Let the efficiencies be

$$M = 4 \text{ units}$$

$$W = 3 \text{ units}$$

$$C = 1 \text{ units}$$

Total units of work required to

$$\text{finish the task} = 24 \times [4(3) + 3(5)$$

$$+ 1(7)] = 816 \text{ units}$$

Number of days taken =

$$\frac{816}{3(4)+3(2)+1(6)} = \frac{816}{24} = 34 \text{ days}$$

Video Solution:



Q17 Text Solution:

Considering it takes n days to complete the work,

X works for = (n-3) days

Y works for = 14 days

Z works for = (n-1) days

$$\text{X's one day of work} = \frac{1}{50}$$

$$\text{Y's one day of work} = \frac{1}{70}$$

$$\text{Z's one day of work} = \frac{1}{90}$$

So

$$\frac{n-3}{50} + \frac{14}{70} + \frac{n-1}{90} = 1$$

$$\Rightarrow 9(n-3) + 90 + 5(n-1) = 450$$

$$\Rightarrow 14n - 32 = 360$$

$$\Rightarrow n = \frac{392}{14} = 28 \text{ days}$$

Video Solution:



Q18 Text Solution:

Let the work done by one man in one day be one unit. Then the total work required for the job is,

$$W = 12 + 12 + \dots (12 \text{ times}) + 11 + 10 + 9 + \dots + 3 + 2 + 1$$

$$= 144 + 11 \times 6 = 210 \text{ units}$$

If one man works on the 1st day, and a new man joins from the 2nd day onwards, then

$$1 + 2 + 3 + \dots + N = 210$$

$$N(N+1)/2 = 210$$

$$N(N+1) = 420 = 20 \times 21$$

$$N = 20$$



Video Solution:**Q19 Text Solution:**

Let us represent Abli, Babli and Chabli by A, B and C respectively for easier understanding of the solution.

Suppose the work done by B in a day is k.

Then the work done in a day by A is 2k while that done by C is 6k. The total work to be done is 75k. Now the work will proceed as follows:

Day 1 (A and B): work done 3k

Day 2 (B and C): work done 7k

Day 3 (A and C): work done 8k

Day 4 (A and B): work done 3k

This pattern will repeat in a cycle of 3 days, and the work done per cycle will be 18k. After 4 such cycles (12 days) the work done will be 72k, at which point the remaining 3k of work will be exactly completed on the 13th day by A and B. In the first 12 days, each of the three will have worked for 8 days, so after the 13th, both A and B would have worked for 9 days in total.

Video Solution:**Q20 Text Solution:**

Let the time both together take to make the furniture be x days.

Time taken by Vikram alone = x + 9 days

Time taken by Dhanush alone = x + 16 days

One day's work when they both work together = Sum of their individual per day work.

$$\frac{1}{x} = \frac{1}{x+16} + \frac{1}{x+9}$$

$$(x+9)(x+16) = x(x+9) + x(x+16)$$

$$\Rightarrow x^2 + 25x + 144 = 2x^2 + 25x$$

$$\Rightarrow x^2 = 144$$

$$x = 12, -12$$

But x cannot be negative, so x = 12.

Time taken by Vikram to make furniture alone = 12 + 9 = 21 days

Alternatively,

We can directly calculate the number of days required for them to complete the work together (t) by the following formula

$$t = \sqrt{t_1 \cdot t_2}$$

$$t = \sqrt{9 \times 16}$$

$$t = \sqrt{144}$$

$$t = 12$$

Therefore, Vikram would require t + 9 i.e. 21 days to finish the work alone.

Video Solution:**Q21 Text Solution:**

Let the work done by a man in a day be a and the work done that by a machine in a day be b, so

If the total work to be done is 1, then

$$\frac{1}{5a+8b} = \left(\frac{40}{100}\right) \left(\frac{1}{4a+3b}\right)$$

$$\Rightarrow 4a + 3b = 2a + 3.2b$$

$$\Rightarrow 2a = 0.2b$$

$$\Rightarrow a = \frac{b}{10}$$

So a man is one – tenth as efficient as a machine. So since 8 machine takes 3 days to finish a job

$$\text{Total work in the job} = 8b \times 3 = 24b$$

So time required to do the same job by 4

$$\text{men} = \frac{24b}{4a} = \frac{6b}{\frac{b}{10}} = 60 \text{ days}$$



Video Solution:**Q22 Text Solution:**

Let the number of days required for Neha to finish the work be x and that of Nisha be y , also let the total number of days required to finish the work when both of them work together be z , so

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = 1$$

Similarly

$$\frac{(z+7)-6}{x} + \frac{(z+7)-9}{y} = 1$$

$$\Rightarrow \frac{z+1}{x} + \frac{z-2}{y} = 1$$

So

$$\frac{z+1}{x} + \frac{z-2}{y} = \frac{z}{x} + \frac{z}{y}$$

$$\Rightarrow \frac{1}{x} = \frac{2}{y}$$

$$\Rightarrow \frac{x}{y} = \frac{1}{2}$$

So required ratio

$$\frac{\frac{1}{x}}{\frac{1}{y}} = \frac{y}{x} = \frac{2}{1}$$

Video Solution:**Q23 Text Solution:**

Total work units required to finish construction = $16 \times 20 = 320$ units

Let the number of days after which 15 workers leave be x , so

$$25(x) + 10(20-x) = 240$$

$$\Rightarrow 25x + 200 - 10x = 320$$

$$\Rightarrow 15x = 120$$

$$\Rightarrow x = 8 \text{ days}$$

So the 15 workers should leave the work after working for 8 days, so they will work till 8th of July, and stop working at the end of the same day.

Video Solution:**Q24 Text Solution:**

Let total work required to dig the well be $120x$ (LCM of 15, 24, 30)

So

$$A+B = 8x$$

$$B+C = 5x$$

$$A+C = 4x$$

Adding first two equations and subtracting the third

$$A+2B+C-A-C = 13-4$$

$$\Rightarrow B = 4.5x$$

$$A = 3.5x$$

$$C = 0.5x$$

Ratio of work done by B to total work done = $B : A+B+C = 4.5x : 8.5x = 9 : 17$

$$\text{So B's share} = \frac{9}{17} \times 680 = \text{Rs } 360$$

Video Solution:**Q25 Text Solution:**

Let the volume of the tank be = LCM(4,6,8,12) = 24 units

$$\text{So efficiency of P} = \frac{24}{4} = 6 \text{ units}$$

$$\text{Efficiency of Q} = \frac{24}{6} = 4 \text{ units}$$



$$\text{Efficiency of R} = -\frac{24}{8} = -3 \text{ units}$$

$$\text{Efficiency of S} = -\frac{24}{12} = -2 \text{ units}$$

In 15 minutes,

$$P \text{ fills} = \frac{6}{4} = \frac{3}{2} \text{ units}$$

$$Q \text{ fills} = \frac{4}{4} = 1 \text{ unit}$$

$$R \text{ empties} = \frac{3}{4} \text{ units}$$

$$S \text{ empties} = \frac{2}{4} = \frac{1}{2} \text{ unit}$$

So all pipes in one hour will fill =

$$\frac{3}{2} - \frac{3}{4} + 1 - \frac{1}{2} = \frac{5}{4} \text{ units}$$

Although, they are filling $\frac{5}{4}$ units every hour, the pipes are first filling and then emptying the tank and hence the tank will get filled to its capacity while being filled by the pipe P or Q, so rather than dividing the capacity of the tank with the rate of filling per hour, we need to observe when the tank gets filled by the pipe, i.e., in the first or the third cycle of the hour. After the first cycle every hour the tank is filled by $\frac{3}{2}$ units and after the third cycle the tank

$$\text{is filled by} = \frac{3}{2} - \frac{3}{4} + 1 = \frac{7}{4} \text{ units}$$

We need to find after how many hours the tank remaining to be filled is less than these two values.

Time required to fill first 22.5 units =

$$\frac{\frac{22.5}{\frac{5}{4}}}{4} = 18 \text{ hours}$$

In the next 15 minutes the pipes will fill another $\frac{3}{2}$ units.

$$\text{So total time required to fill the tank} = (18 + 0.25) \times 60 = 1095 \text{ minutes}$$

Video Solution:



Q26 Text Solution:

Let David need x hours to finish the job alone.

Then, John needs $(x + 9)$ hours to finish the job alone.

David, Rudra and John together complete the job in $(x + 9 - 8) = (x + 1)$ hours

Rudra alone can destroy the work in 6 hours.

Therefore, we have

$$\frac{1}{x} + \frac{1}{x+9} - \frac{1}{6} = \frac{1}{x+1}$$

$$\frac{2x+9}{x(x+9)} = \frac{6+x+1}{6(x+1)}$$

$$\frac{2x+9}{(x^2+9x)} = \frac{7+x}{6(x+1)}$$

$$12x^2 + 12x + 54x + 54 = 7x^2 + x^3 + 63x + 9x^2$$

$$x^3 + 4x^2 - 3x - 54 = 0$$

$$x^2(x - 3) + 7x(x - 3) + 18(x - 3) = 0$$

$$(x - 3)(x^2 + 7x + 18) = 0$$

So, $x = 3$ as other two solutions are imaginary for being the discriminant less than zero.

Hence, together David, Rudra and John finish the entire current work of painting the house in $(3 + 1) = 4$ hours.

Therefore, they together will finish the painting work of a house **twice** as large as the first one in $4 \times 2 = 8$ hours

Video Solution:



Q27 Text Solution:

In every 3 days, all 3 of them are working for 2 days only, which means all of them work together for 48 days to finish the work, so if the efficiencies of Ravi, Sanjay and Priya is r , s and p , then

$$48r + 48s + 48p = 1$$

Given Ravi alone can complete the work in 96 days, so

$$r = \frac{1}{96}$$

Also given

$$s = 1.5p$$



[Android App](#)

| [iOS App](#)

| [PW Website](#)

So

$$\frac{48}{96} + 72p + 48p = 1$$

$$\Rightarrow \frac{1}{2} + 120p = 1$$

$$\Rightarrow p = \frac{1}{240}$$

So

$$s = 1.5 \times \frac{1}{240} = \frac{1}{160}$$

Time required by Sanjay to finish half the work

$$= \frac{\frac{1}{2}}{\frac{1}{160}} = 80 \text{ days}$$

Video Solution:



Q28 Text Solution:

Given that,

6 Formica and 10 Dorylus ants together can complete the dragging work in 6 minutes and one Formica ant can complete the dragging work alone in 1 hour, i.e., in 60 minutes.

So, 1 Formica ant completes $1/60$ units of work in 1 minute.

i.e., 6 Formica ants complete $6/60 = 1/10$ units of work in 1 minute.

Now, 6 Formica and 10 Dorylus ants together can complete the work in 6 minutes.

So, their combined work rate = $1/6$ units per minute.

Let the work rate of 1 Dorylus ant be x units per minute,

Then, the Work rate of 10 Dorylus ants = $10x$ units per minute.

Therefore, we have

$$1/10 + 10x = 1/6$$

$$10x = 1/6 - 1/10$$

$$x = 1/150$$

i.e., 1 Dorylus ant can complete the work alone in 150 minutes.

Video Solution:



Q29 Text Solution:

Let pipe C takes x hours to fill the tank alone.

So, Pipe B takes $(x + 3)$ hours to fill the tank alone.

Then, pipe A takes $(x + 3 + 9) = (x + 12)$ hours to fill the tank alone.

Thus, pipe C will take = $\sqrt{3 \times 12} = 6$ hours alone to fill the tank.

i.e., pipe A takes = $(6 + 12) = 18$ hours to fill the tank alone.

and pipe B takes = $(6 + 3) = 9$ hours to fill the tank alone.

Let the leak take y hours to drain all the water out of the tank alone.

It is given that all the pipes alone with the leak take 24 hours to fill the tank.

So, total capacity of the tank = LCM (18, 9, 6, 24) = 72 units

So, the efficiency of pipe A = 4

The efficiency of pipe B = 8

The efficiency of pipe C = 12

The efficiency of all the pipes and the leak = 3

So, the efficiency of the leak = $3 - (4 + 8 + 12) = -21$

Thus, it would take = $\frac{72}{21} \approx 3$ hours to empty the entire tank by the leak.

Video Solution:



Q30 Text Solution:

Let's take the total work to be done be 100 units, also let the initial number of workers be x and their work efficiency be y units/day, so in first 30 days x workers working at a efficiency of y units/day will finish 30% of work

$$30 \times x \times y = \frac{30}{100} \times 100$$

$$\Rightarrow xy = 1 \text{ unit/day}$$

Also, when 10 more workers joined

$$15 \times (x + 10) \times y = \frac{20}{100} \times 100$$

$$\Rightarrow xy + 10y = \frac{4}{3}$$

$$\Rightarrow 1 + 10y = \frac{4}{3}$$

$$\Rightarrow 10y = \frac{1}{3}$$

$$\Rightarrow y = \frac{1}{30} \text{ unit/day}$$

So the efficiency of 1 worker is $\frac{1}{30}$ units/day.

So

$$xy = 1$$

$$\Rightarrow x = \frac{1}{y} = 30 \text{ workers}$$

In the last 30 days total $x+10+n$, which means $40+n$ workers are going to work to complete the remaining 50% of work at the efficiency of

$$\frac{1}{30} \text{ units/day}$$

So

$$(40 + n) \times 30 \times \frac{1}{30} = \frac{50}{100} \times 100$$

$$\Rightarrow n + 40 = 50$$

$$\Rightarrow n = 10$$

Video Solution:



[Android App](#)

| [iOS App](#)

| [PW Website](#)